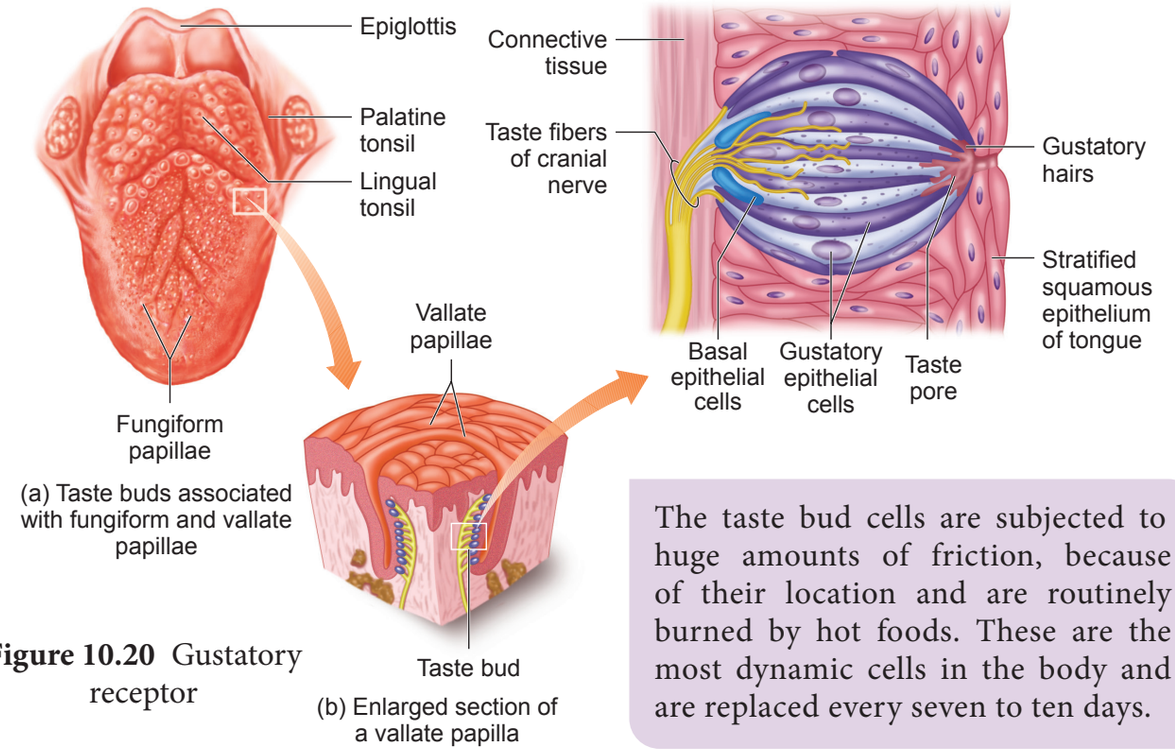


**Figure 10.20** Gustatory receptor

The taste bud cells are subjected to huge amounts of friction, because of their location and are routinely burned by hot foods. These are the most dynamic cells in the body and are replaced every seven to ten days.

lobe of the brain for identification of smell and the limbic system for the emotional responses to odour.

**Gustatory receptor:** The sense of taste is considered to be the most pleasurable of all senses. The tongue is provided with many small projections called **papillae** which give the tongue an abrasive feel. Taste buds are located mainly on the papillae which are scattered over the entire tongue surface. Most taste buds are seen on the tongue (Figure 10.20) few are scattered on the soft palate, inner surface of the cheeks, pharynx and epiglottis of the larynx. Taste buds are flask-shaped and consist of 50 – 100 epithelial cells of two major types.

**Gustatory epithelial cells** (taste cells) and **Basal epithelial cells** (Repairing cells) Long microvilli called **gustatory hairs** project from the tip of the gustatory cells and extends through a taste pore to the surface of the epithelium where they are bathed by saliva. Gustatory hairs are the sensitive portion of the gustatory cells and they have sensory dendrites which send the signal to the brain. The basal cells that act as stem cells, divide and differentiate into new gustatory cells (Figure 10.20).

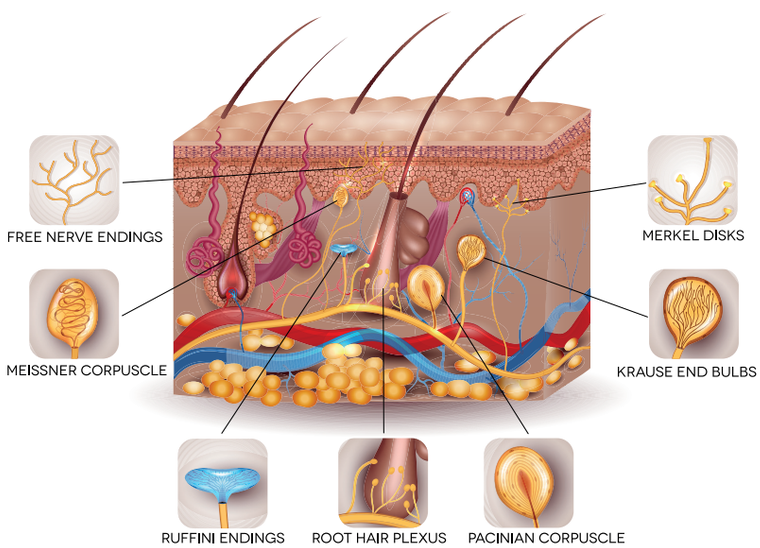


**Figure 10.21** Skin receptors

### Skin-Sense of touch

Skin is the sensory organ of touch and is also the largest sense organ. This



XI Std Zoology Chapter-10 EM.indd 196 30/12/2022 13:56:01